Module Code & Module Title

CS6P05NP & Final Year Project (Proposal)

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Assessment Weightage & Type

5% Proposal Report

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I acknowledge that my proposal report must be submitted online through MY SECOND TEACHER on the appropriate module page before the assignment deadline to be accepted and assessed. I know that late submissions will be considered non-submissions, resulting in a zero mark.

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1. INTRODUCTION

The title of the project, “SkillXchange: A Web-Based Skill Exchange Platform,” represents an innovative approach to learning and collaboration. This project aims to build an online system where people can exchange skills directly without the use of money. In today’s modern world, many individuals have valuable talents such as photography, music, programming, or designing, but they lack the funds to pay for professional courses. The SkillXchange platform acts as a meeting point for learners and teachers, allowing them to share their skills in a fair, transparent, and rewarding environment.

The system is designed as a responsive web application using the Laravel framework with PHP as the backend, HTML, CSS, and JavaScript for the frontend, and MySQL for the database. The platform consists of three main sections: the Admin Dashboard**,** User Panel, and Skill Exchange Area, each serving a specific purpose to ensure smooth functionality. The admin monitors all activities, manages security, and ensures the system is functioning properly, while users can create profiles, list their skills, send barter requests, and rate other users after successful sessions.

The SkillXchange platform is more than just a website; it is a community-driven initiative that fosters personal development, self-learning, and social interaction through skill sharing. It integrates gamification, AI-based matchmaking, and premium membership features, making the entire experience interactive, motivating, and efficient.

1.1 Problem Scenario

In today’s fast-paced digital era, learning new skills or accessing professional services often requires financial capability. Many individuals, particularly students, freelancers, or those unemployed, possess skills they can share but cannot afford to acquire new learning opportunities. Existing platforms focus mostly on paid courses, freelance hiring, or volunteering, but lack a proper structure for skill exchange. This creates a missed opportunity for people to collaborate and grow without financial pressure. Additionally, there is no centralized system in Nepal that provides a trustworthy, fair barter mechanism for skill exchange, making it difficult for users to find matching partners and maintain transaction transparency.

1.2 The Project as a Solution

The Skill Barter System offers a practical solution by creating a digital platform where users can register, list their skills, and search for other users willing to exchange them. For instance, a person skilled in graphic design can exchange their service with someone offering programming lessons. The platform ensures secure authentication, mutual agreement verification, and feedback mechanisms to maintain trust. Users can browse listings, propose trades, and communicate through an integrated messaging system. This project contributes to skill development, community collaboration, and economic inclusion, empowering individuals to grow both personally and professionally.

1.3 Related Projects

Several existing systems have inspired the development of the Skill Barter System. Platforms like LinkedIn Learning, Fiverr, and Skillshare focus on paid services, while platforms such as Simbi promote non-monetary exchange. However, these platforms often lack localized features suitable for Nepalese users, such as language options, community-based reputation systems, and easy barter matching. The Skill Barter System adopts the best features from these global models and tailors them for the local context.

1.3.1 LinkedIn Learning

LinkedIn Learning is a well-known platform that provides thousands of online video courses taught by professionals. It primarily focuses on professional skill development and career advancement. Users can complete courses and showcase their achievements on their LinkedIn profiles. However, the system is fully monetary-based, meaning users must pay for most courses or subscribe to premium plans. It does not allow direct interaction or skill exchange between users. In contrast, SkillXchange focuses on free, peer-to-peer skill exchange, where users teach and learn from one another without any cost, fostering a sense of community and cooperation.

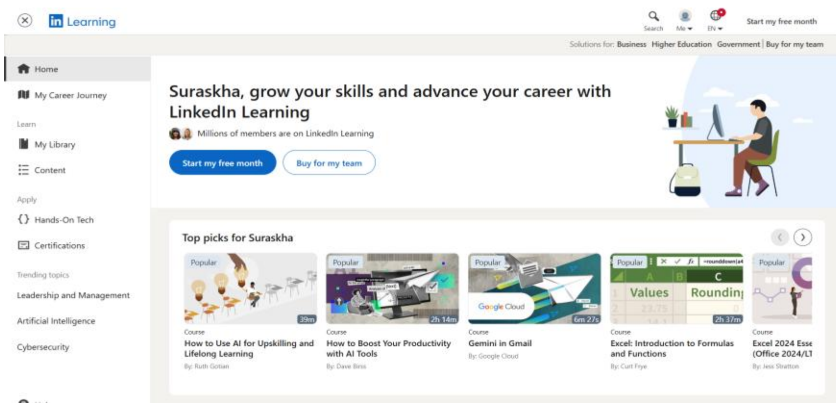

figure 1: LinkedIn Learning's Home Page

1.3.2 Skillshare

SkillShare is another global learning platform that allows experts to create and share video lessons in creative fields like design, photography, and writing. Users must pay for a subscription to access all classes, which limits opportunities for those who cannot afford paid learning. SkillShare is focused on one-way learning from instructors to students, without mutual exchange or collaboration. The SkillBarter platform, on the other hand, emphasizes two-way learning, where every user can be both a teacher and a learner. It uses an AI-based matching system to connect users who have complementary skills and includes a gamified system with badges and points to make learning fun and motivating.

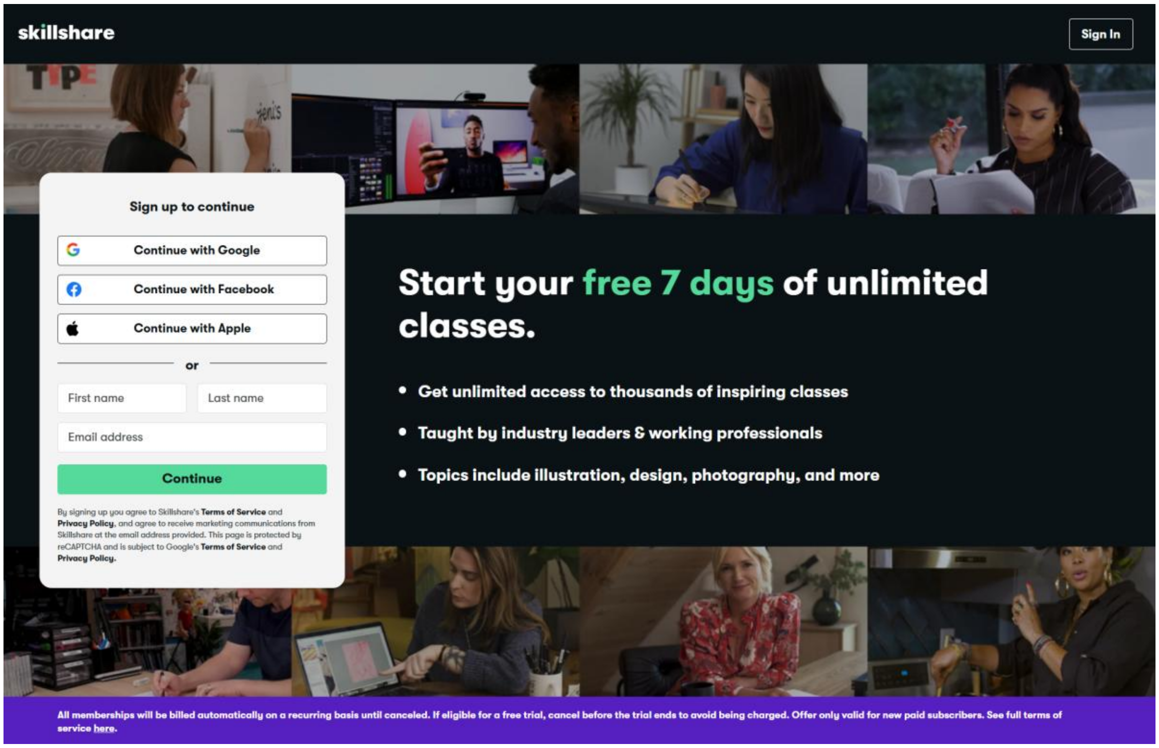

figure 2: Skillshare

1.3.3 Simbi

Simbi is one of the few platforms that allow users to trade services without using money. It works on a credit-based system where users earn credits by offering their skills and spending them to receive services from others. Although Simbi promotes non-monetary exchange, it lacks AI matchmaking, gamification features, and a localized design for users in Nepal. Simbi also does not provide built-in communication or scheduling tools. The SkillXchange system addresses these limitations by introducing an AI skill-matching algorithm, in-app chat and calendar scheduling, and Premium Membership features for users seeking an enhanced experience.

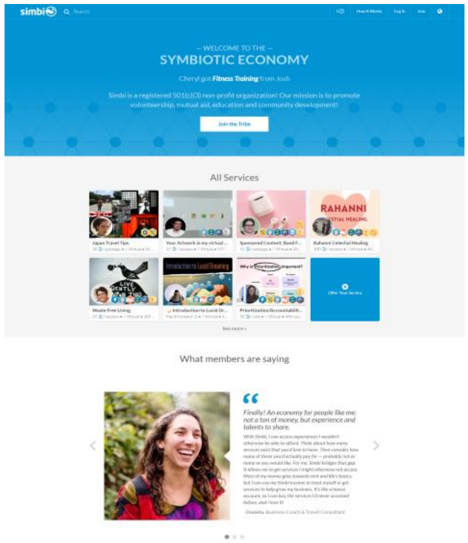

figure 3: Simbi

1.4 Comparison

The comparison below highlights the distinctions between similar systems and the proposed skill barter system.

Table 1: Compare my project to another project

Feature	LinkedIn Learning	SkillShare	Simbi	SkillXchange
Skill Exchange	-	-	✓	✓
Monetary Requirement	✓	✓	-	-
Rating & Feedback	✓	✓	✓	✓
Scheduling & Chat	-	-	-	✓
Gamification	-	-	-	✓
AI-Based Matching	-	-	-	✓
Premium Membership	✓	✓	-	✓
Peer-to-Peer learning	-	-	✓	✓

2. AIMS AND OBJECTIVES

2.1 Aims

The aim of this project is to develop a web-based platform that enables people to exchange their skills fairly and efficiently without using money. The system will promote self-learning, communication, and community engagement through secure authentication, AI-based matchmaking, gamification, and feedback systems.

2.2 Objectives

- To create a secure login and registration system for users.
- To implement a skill management module where users can add “skills to teach” and “skills to learn.”
- To design an efficient matching algorithm for skill barter requests.
- To provide scheduling and chat features for session management.
- To develop a responsive interface using HTML, CSS, and JavaScript.
- To integrate a feedback and rating system for user reliability.
- To integrate a payment gateway for Premium Membership.
- To use the Incremental Software Development Model for phased delivery.

3. EXPECTED OUTCOMES AND DELIVERABLES

3.1 Expected Outcomes

- A fully functional web platform that supports skill exchange without monetary transactions.
- AI-based partner matching for fair and efficient skill barter.
- Improved user engagement through gamification and rewards.
- Premium users enjoy enhanced features like AI suggestions and resource uploads.
- A user-friendly interface promoting fair trade and interaction.
- Community development through collaboration and shared learning.

3.2 Deliverables

- Project proposal report
- System prototype and documentation
- Final project report and presentation
- Source code with user manual

4. PROJECT RISKS, THREATS, AND CONTINGENCY PLANS

Table 2: Project Risks, Threats, and Contingency Plans

Problems	Technical Issues	Security Breaches	Health Issues	Schedule Delays
Risk	Bugs or system crash	Unauthorized data access	Fatigue or illness	Missed milestones
Threat	Project delay	Loss of trust	Reduced productivity	Deadline extension
Plan	Regular code review and testing	Apply encryption and strong authentication	Maintain work-life balance	Adjust timeline and review progress weekly

5. METHODOLOGY

The development of the Skill Barter System will follow the Incremental Software Development Methodology. This model divides the project into smaller, manageable increments that are developed, tested, and refined in successive cycles. Each increment adds functionality to the system, ensuring continuous progress and early feedback. This approach suits the project well since it allows flexibility in requirements and iterative improvements based on user testing.

5.1 Model Overview

5.1.1 Waterfall Model

The Waterfall Model, introduced by Winston W. Royce in 1970, follows a simple step-by-step development process. The phases include Requirement Analysis, System Design, Implementation, Testing, Deployment, and Maintenance. Each phase is completed before moving to the next one. This model is best for projects where requirements are clearly defined from the beginning and do not change frequently.

Advantages:

- Fixed timeline with well-defined phases.
- Easy to manage since each phase has clear deliverables.
- Good for small projects with predictable requirements.

Disadvantages:

- It is difficult to handle unexpected changes during development.
- Testing happens late, which may increase cost if major errors appear.
- Less flexibility and limited user feedback during early stages.

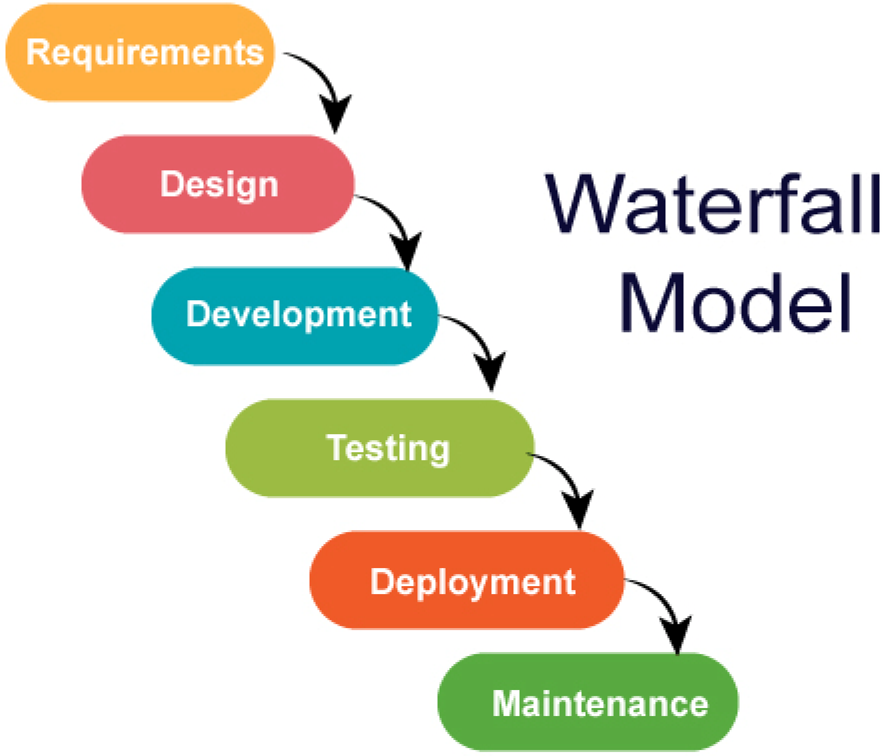

figure 4: Watterfall model

5.1.2 Prototype Model

The Prototype Model focuses on building a quick prototype (or demo version) of the system before the final one is developed. This helps users understand what the system will look like and gives feedback early. The cycle includes Requirement Gathering, Quick Design, Prototype Development, Testing, and Refinement until the system meets user expectations.

Advantage:

- Encourages user involvement and feedback.
- Helps visualize the system early in development.
- Increase customer satisfaction because users see progress.

Disadvantages:

- Increase time and cost due to repeated refinements.
- Create confusion between the prototype and the final system.
- Lead to incomplete documentation.

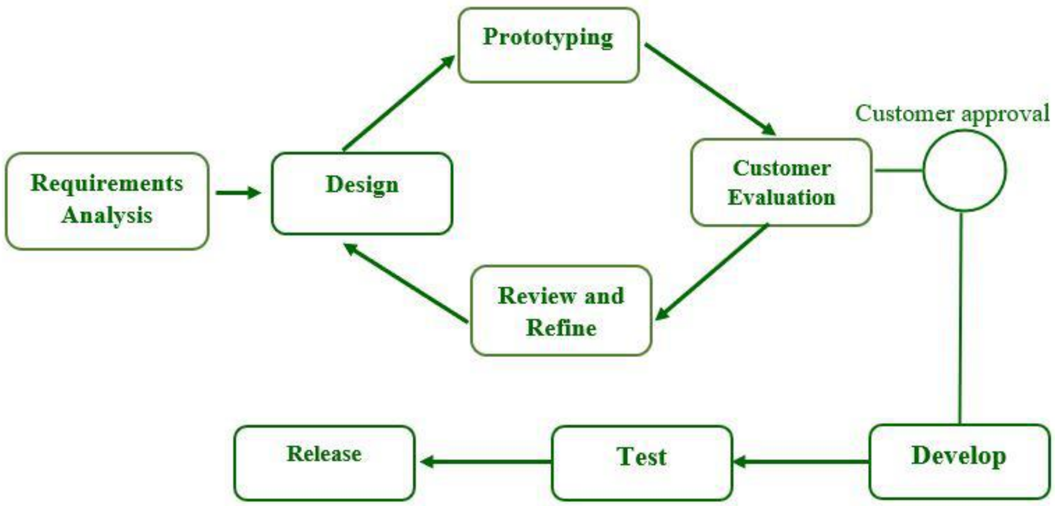

figure 5: Prototype model

5.1.3 Spiral Model

The Spiral Model combines features of both the Waterfall and Prototype models. It is used for large or complex projects that need continuous improvement and risk management. The process repeats in cycles called *spirals*, where each loop includes Planning, Risk Analysis, Engineering, Testing, and Evaluation.

Advantages:

- Focus on risk management and problem-solving.
- Suitable for complex and long-term projects.
- Allows continuous refinement and customer input.

Disadvantages:

- Costly and time-consuming for small projects.
- Hard to manage due to complexity.
- Need experienced developers for risk assessment.

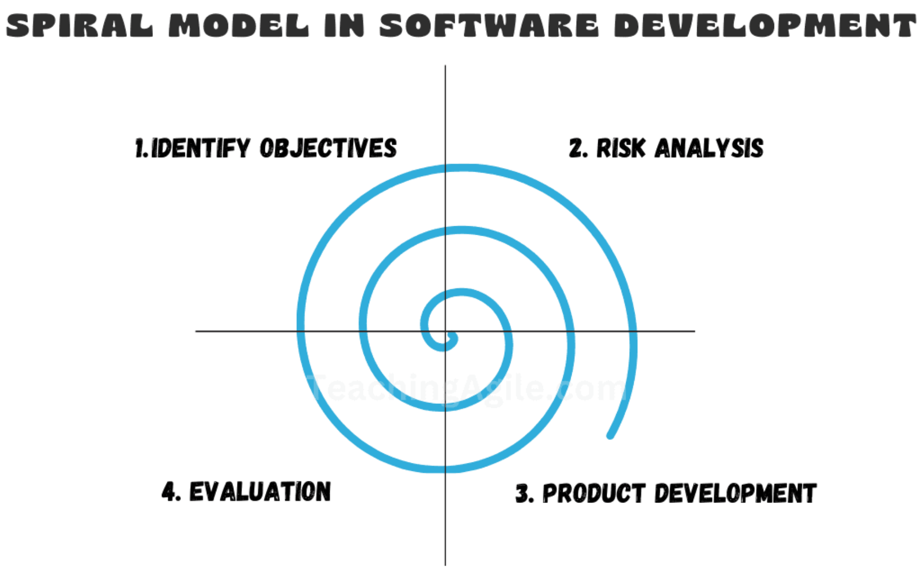

figure 6: Spiral model

5.2 Selected Model

The incremental model in software engineering divides a project into smaller modules, or increments, developed and integrated gradually. This iterative approach enables frequent testing, quick feedback, and early detection (Sachan, 2024).

The Incremental Model is a great fit for this project because it breaks the work into smaller steps, making development and integration easier. Each part is tested and improved before moving on, which helps create a more stable and unified system. This method also allows for regular feedback and makes it easier to adjust to changes, ensuring the final product meets user needs and is strong and reliable.

Advantages of the Incremental Model:

- Phased Development: Divides work into smaller and clearer stages.
- Early Integration: Delivers working parts early for testing.
- Reduced Risk: Errors can be found and fixed early.
- User Feedback: Feedback collected after each increment improves quality.
- Steady Progress: Development continues smoothly without delays.
- Adaptability: Easier to adjust features based on user needs or new ideas.

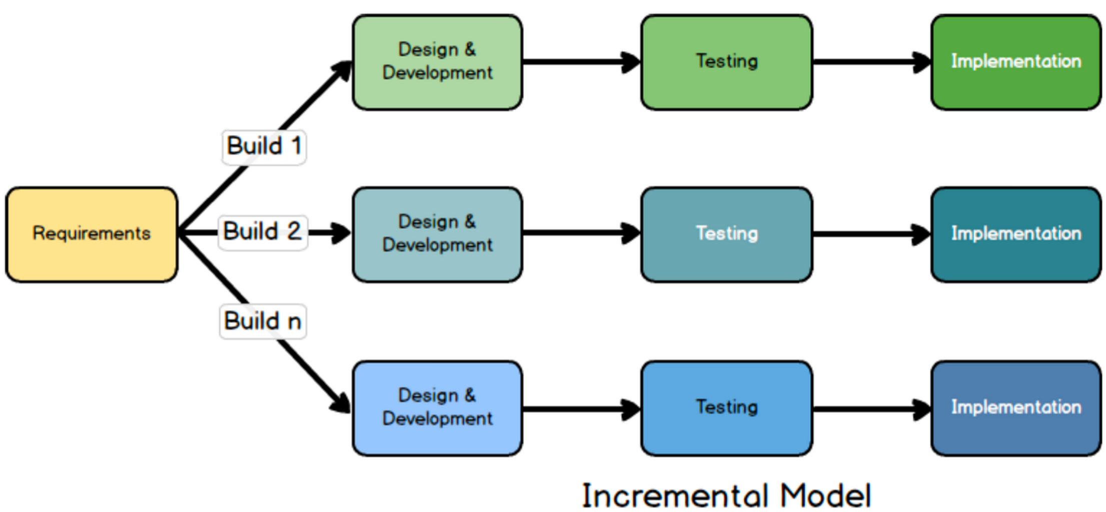

figure 7: Incremental Model

6. RESOURCE REQUIREMENTS

Database: MySQL

XAMPP is an open-source local development environment created by Apache Friends. It includes Apache HTTP Server, MySQL Database, and interpreters for PHP and Perl, making it ideal for web development. MySQL will be used to manage all user data, skill records, and barter requests securely. XAMPP allows for easy setup and local testing before deployment (DSouza, 2024).

figure 8:Xampp

- **Framework: Laravel**

Laravel is an open-source PHP framework that follows the Model-View-Controller (MVC) pattern. It provides built-in tools for routing, authentication, and database management, ensuring structured and efficient coding. For this project, Laravel will be used to build the core functionality of the Skill Barter System, manage data flow between the front end and back end, and maintain system security (tutorialspoint, 2024).

figure 9: Laravel

- **Frontend: HTML, CSS, and JavaScript**

  HTML defines the structure of the webpages, CSS enhances the layout and design, and JavaScript adds interactivity and logic to improve user experience. Together, they ensure the Skill Barter System is visually appealing, responsive, and easy to navigate on both desktop and mobile devices.

figure 10: HTML,CSS & JS

- **Backend: PHP**

  PHP is a powerful scripting language used for handling backend processes such as user authentication, barter request handling, and database communication. It ensures smooth operation between client and server, making the platform secure and dynamic (WillDom, 2023).

figure 11: PHP

- **Text Editor: Visual Studio Code**

  Visual Studio Code is a free, open-source code editor that supports multiple programming languages. It offers features like syntax highlighting, debugging tools, and real-time collaboration, which help streamline the development process for the Skill Barter System (UCHENDU, 2021).

figure 12: VS Code

- **Documentation: Microsoft Word**

  Microsoft Word will be used to prepare reports, documentation, and proposals for the Skill Barter System. It allows for organized formatting, editing, and integration of figures and tables, ensuring professional presentation of written materials.

figure 13: MS word 365

- **Design Tools: Figma and Draw.io**

  Draw.io (also known as Diagrams.net) will be used to create system flowcharts, ER diagrams, and UML models. It supports cloud storage and data privacy, which makes it convenient for diagramming project designs (Edirisinghe, 2023).

  Figma will be used for UI/UX design. It is a collaborative online tool that allows for wireframing and interface design, helping create a modern and user-friendly look for the Skill Barter System (Purdila, 2023).

figure 14: Figma

figure 15: Draw.io

7. WORK BREAKDOWN STRUCTURE (WBS)

The following breakdown represents the major phases of development using the Incremental Model:

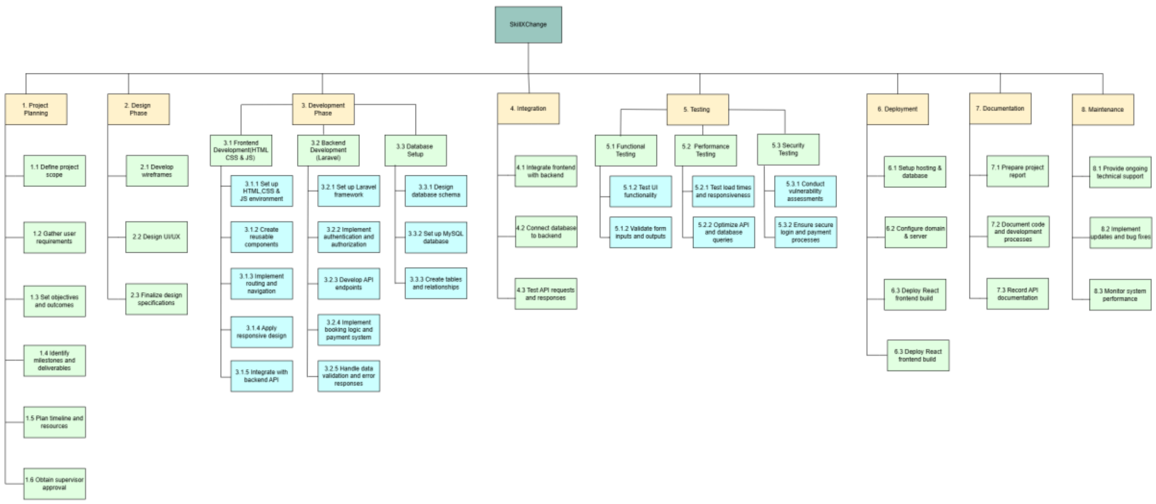

figure 16: Work Breakdown Structure

8. MILESTONES

Table 3: Milestones

Milestones	Activities	Dates
1	Finalize Topic	25 July 2025 to 08 Aug 2025
2	Topic Selection	09 Aug 2025 to 14 Aug 2025
3	Gather Requirements	15 Aug 2025 to 05 Sep 2025
4	Design Approval	06 Sep 2025 to 10 Oct 2025
5	Proposal Report Submission	11 Oct 2025 to 14 Nov 2025
6	Frontend Development	15 Nov 2025 to 10 Dec 2026
7	Backend Development	11 Dec 2025 to 10 Jan 2026
8	Database Setup	11 Jan 2026 to 10 Feb 2026
9	Integration Testing	12 Feb 2026 to 25 Feb 2026
10	Release Alpha Testing	26 Feb 2026 to 03 Mar 2026
11	Beta Testing	04 Mar 2026 to 15 Mar 2026
12	User Feedback Collection	16 Mar 2026 to 25 Mar 2026
13	Refine Features	26 Mar 2026 to 30 Mar 2026
14	Final Testing	01 Apr 2026 to 08 Apr 2026
15	Deployment	09 Apr 2026 to 15 Apr 2026
16	Submit Documentation	16 Apr 2026 to 26 Apr 2026
17	Review Project	27 Apr 2026 to 02 May 2026
18	Submit Final Report	03 May 2026
19	Prepare Presentation	04 May 2026
20	Implement Feedback	05 May 2026
21	Close Project	06 May 2026
22	Final Evaluation	07 May 2026

9. PROJECT GANTT CHART

The Gantt Chart below represents the timeline of the Skill Barter System project across the current academic term. Each bar corresponds to the time allocated for major project milestones.

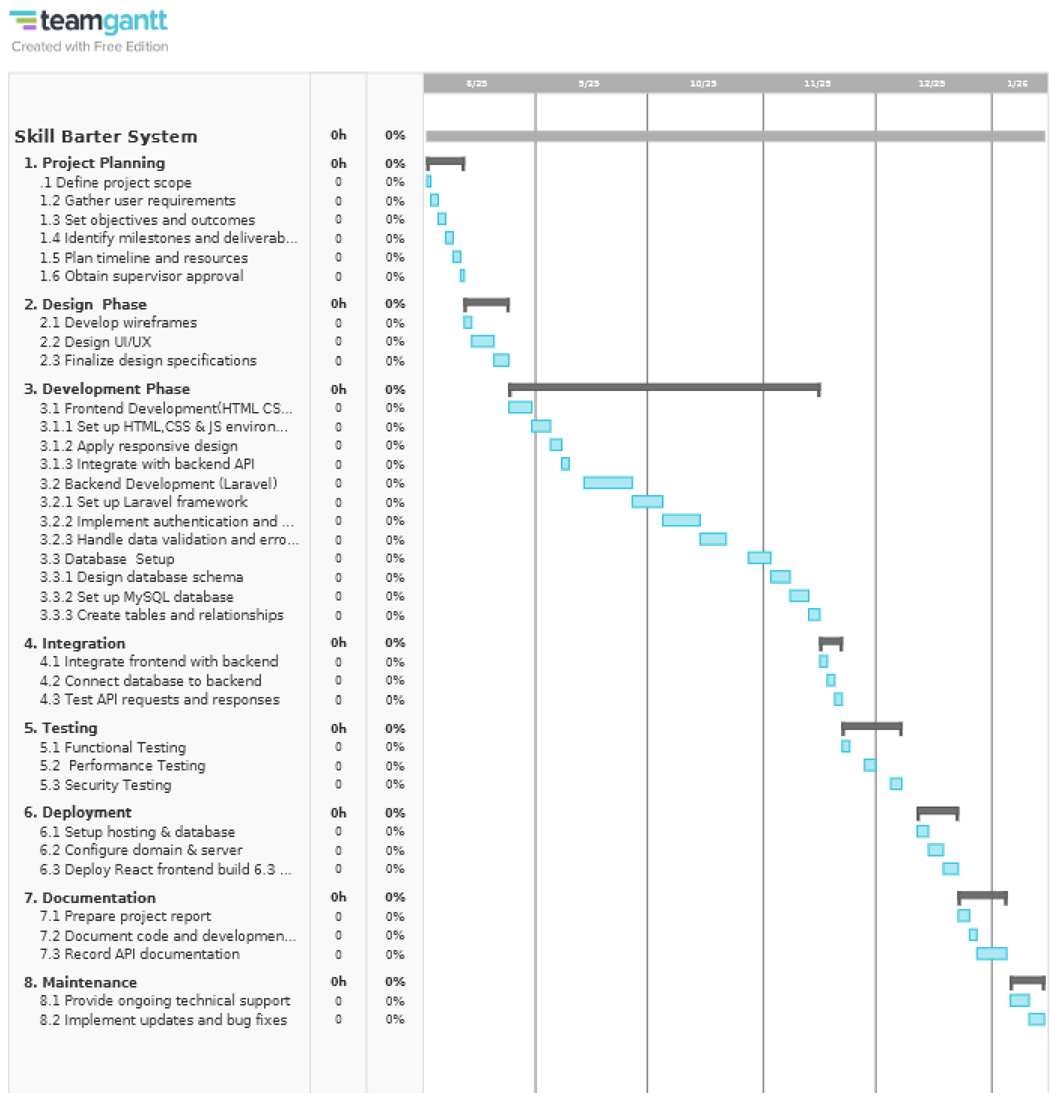

figure 17: Gantt chart

10. CONCLUSION

The SkillXChange project aims to build an online platform where people can share and learn new skills through direct exchange, rather than money. It helps users connect with others who have complementary skills through AI-based matching and offers features such as chatting, scheduling, feedback, and ratings to make learning more interactive. The system also includes gamified elements such as points, badges, and leaderboards to encourage user participation. For those who want advanced options, a Premium Membership provides extra benefits like unlimited barter requests, AI skill suggestions, ad-free browsing, and unlimited resource sharing, giving users more flexibility and control over their learning experience. Overall, the project focuses on creating a practical, secure, and user-friendly platform that encourages knowledge sharing, teamwork, and personal growth, helping users learn and teach skills in an interactive digital community.

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